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Received: 2 February 2026 | Accepted: 8 September 2026

DOI: 10.1111/bjet.70089

ORIGINAL ARTICLE

British Journal of Educational Technology BERA

Many hands make light work? Individual and collaborative interaction with a GenAI-powered multi-agent system: Effects on learning performance and critical thinking

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Funding information
2026 Key Project of the Zhejiang Provincial Philosophy and Social Sciences Planning Programme, Grant/Award Number: 26NDJC005Z; 2024 Ministry of Education Country and Area Studies Project, Grant/Award Number: 2024-N02; 2024 Zhejiang Provincial Education Science Planning Project, Grant/Award Number: 2024SCG247

Abstract

The rapid proliferation of generative artificial intelligence (GenAI) has made fostering learners' critical thinking an increasingly pressing educational challenge. While single-agent GenAI applications are prone to limited perspective diversity and echo-chamber effects, multi-agent systems (MAS) offer a promising alternative by enabling multi-perspective interactions. Meanwhile, the pedagogical potential of collaborative human–GenAI interaction, particularly when peers jointly engage with MAS to promote epistemic diversity, remains underexplored. Moreover, little is known about how critical thinking unfolds within such interactions at a process-oriented level. This study investigates how different configurations of peer collaboration and GenAI-powered multi-agent support are associated with learning performance, critical thinking awareness and interaction processes. A quasi-experimental study was conducted with 87 undergraduate students enrolled in an instructional design course. Participants were assigned to a peer collaboration condition without AI support, an individual human–MAS interaction condition or a collaborative human–MAS interaction condition. Learning performance was assessed using knowledge tests and design projects, critical thinking awareness was measured via a questionnaire and critical thinking processes were examined through lag sequential analysis and epistemic network analysis. The results indicate that collaborative interaction

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with the MAS corresponded to higher project performance, whereas individual interaction was associated with greater gains in critical thinking awareness. Process analyses further suggest distinct cognitive trajectories across configurations, with individual interaction characterised by more focused analytical–evaluative processing and collaborative interaction involving more distributed and integrative reasoning. These findings highlight the importance of aligning MAS design with appropriate human–AI collaboration configurations to better facilitate critical thinking in GenAI-supported learning environments.

KEYWORDS

critical thinking, generative artificial intelligence, human–AI collaboration, interaction processes, multi-agent systems

INTRODUCTION

As generative artificial intelligence (GenAI) is increasingly integrated into learning environments, the importance of critical thinking has become more imperative than ever (Cukurova, 2025; Yan et al., 2024). This heightened importance arises because GenAI not only offers promising affordances for supporting academic learning, but also paradoxically introduces recurring risks, including hallucinated outputs and learners' overreliance on AI-generated responses (Yan et al., 2024). Addressing these risks requires more intentional human–GenAI collaboration designs that support learners in critically engaging with AI-generated content, evaluating information quality and regulating their reliance on intelligent systems (Hou et al., 2025). However, most current implementations of GenAI in learning settings have relied on a single GenAI agent serving a specific role (Hao et al., 2026). Such monolithic interaction structures are prone to 'echo-chamber' effects (Jacob et al., 2025), which foster learner passivity at the expense of critical engagement and academic achievement.

Multi-agent systems (MAS) have emerged as a potential architectural response to the limitations of single-agent GenAI applications. By orchestrating multiple specialised virtual agents, MAS provide diversified and role-differentiated cognitive support, enabling learners to engage with problems from multiple perspectives (Hao et al., 2026; Zheng et al., 2026). Recent empirical evidence suggests that these multi-role environments provide multidimensional informational support that enhances learning achievement (Zheng et al., 2026) and also catalyse higher-order thinking, supporting structured reasoning, systematic problem-solving and perspective-taking (Shi et al., 2025).

Nevertheless, the pedagogical benefits of MAS are contingent not on the mere presence of multiple agents, but on how learners are configured to interact with them, either individually or collaboratively (Chu & Li, 2025). While individual interaction with a MAS may foster intense self-regulated reasoning, collaborative configurations involving human peers introduce layers of social negotiation, exposure to multiple perspectives and shared epistemic agency (Tai & Chen, 2024; Zou et al., 2023). These differing interactional configurations are therefore likely to be associated with not only learning performance but also the underlying cognitive processes through which critical thinking develops.

Despite these promising propositions, empirical evidence comparing the cognitive impacts of different human–MAS configurations remains sparse. Prior research has often

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Practitioner notes

What is already known about this topic:

- Generative artificial intelligence (GenAI) is increasingly being incorporated into educational settings, raising ongoing interest in how AI-supported learning environments can engage learners in higher-order cognitive processes, including critical thinking.
- Many existing AI-supported learning environments rely on single-agent systems, which may unintentionally encourage answer-seeking behaviours and reinforce echo-chamber effects.
- Multi-agent systems (MAS) provide role-differentiated support and exposure to diverse perspectives, and have been suggested as a promising approach for supporting higher-order thinking in learning.

What this paper adds:

- By developing a GenAI-supported multi-agent system, this study empirically compares learning outcomes across peer collaboration without AI support, individual interaction with the MAS, and collaborative interaction with the MAS, showing that individual interaction is linked to higher critical thinking awareness gains, whereas collaborative interaction corresponds to stronger performance on complex design-oriented tasks.
- Through the integration of lag sequential analysis and epistemic network analysis, the study reveals how overt interaction behaviours and implicit cognitive structures jointly reflect critical thinking processes in multi-agent learning environments.
- The findings provide initial process-level insights suggesting that multi-agent systems may function not merely as information providers, but as reflective and socially mediating supports that facilitate how learners regulate, negotiate and integrate ideas during learning.

Implications for practice and/or policy:

- Educators can deliberately align human–GenAI interaction configurations with specific instructional goals, as individual interaction with a multi-agent system may be more conducive to learners' critical thinking awareness, and collaborative interaction may be better suited for complex design and problem-solving tasks.
- Instructional designers are encouraged to embed explicit reflective checkpoints or targeted critical thinking scaffolds within collaborative human–MAS interaction settings to mitigate reduced opportunities for individual metacognitive monitoring.

examined peer collaboration and AI-supported learning separately (Li, Hwang, et al., 2025; Tai & Chen, 2024; Xu et al., 2023), but less is known about how these elements function across different instructional configurations within the same authentic educational context. In particular, the interaction patterns through which critical thinking is enacted across individual and collaborative human–MAS arrangements are still underexplored (Ba, Yang, et al., 2025; Zheng et al., 2026), which prevents educators from fully leveraging MAS to support learning performance and critical thinking development. Against this backdrop, the present study compares learning performance and critical thinking awareness across peer collaborative learning without AI support and two forms of human–GenAI interaction supported by the MAS. It also explores the patterns of critical thinking processes that emerge during

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human–GenAI interaction using lag sequential analysis and epistemic network analysis. Specifically, the study addresses the following research questions (RQs):

RQ1. How does learning performance differ across peer collaboration without AI support, individual interaction with the MAS and collaborative interaction with the MAS?

RQ2. How does critical thinking awareness differ across peer collaboration without AI support, individual interaction with the MAS and collaborative interaction with the MAS?

RQ3. How do critical thinking interaction patterns differ across individual and collaborative human–GenAI configurations?

LITERATURE REVIEW

GenAI multi-agent systems and critical thinking

Critical thinking is widely conceptualised as a process-oriented form of judgement that is fundamentally shaped through interaction (Ennis, 1985; Vygotsky, 1978). It refers to the ability to make reasonable judgements about what to do and what to believe (Ennis, 1985), involving processes such as recognising problems, understanding information, analysing arguments, evaluating evidence and generating informed solutions (Facione, 1990; Murphy, 2004). From a social constructivist perspective, higher-order cognitive processes are constructed through interaction, externalisation and meaning negotiation with others (Vygotsky, 1978). Through interaction, learners are encouraged to articulate their reasoning, examine underlying assumptions and refine judgements, thereby supporting the development of critical thinking.

Within this interactional perspective, GenAI has been explored as a potential cognitive partner for supporting critical thinking (Guo et al., 2026). Prior work has highlighted its use in providing feedback, prompting reflection and supporting engagement with complex tasks (Lu & Ba, 2025). Empirical studies further report improvements in learning performance and critical thinking in GenAI-supported environments, often associated with real-time feedback and analytical guidance (Guo et al., 2026; Li, Hwang, et al., 2025). However, interaction with a single GenAI agent also presents recurring challenges. Prior research has identified 'echo-chamber' effects in traditional single-agent interactions, whereby confirmation bias and limited perspective diversity may reduce opportunities for genuine cognitive engagement (Jacob et al., 2025). In such contexts, learners may be inadvertently encouraged towards passive acceptance rather than the critical scepticism required for higher-order cognition.

To address these limitations, multi-agent systems (MAS) have been proposed as an architectural response to the constraints of single-agent GenAI interactions. Unlike monolithic GenAI models, MAS orchestrate multiple specialised agents that perform differentiated roles, thereby fostering a form of distributed inquiry resembling a 'community of inquiry' (Hao et al., 2026). By simulating multi-perspective dialogue and complex problem-solving scenarios, MAS-supported environments can create opportunities for learners to encounter competing viewpoints, evaluate alternative claims and justify their reasoning (Zheng et al., 2026). For instance, an experimental study demonstrated that multi-agent structured feedback, diversified roles and immersive scenarios can significantly enhance law students' problem-solving skills, critical thinking, creativity and perspective-taking (Shi et al., 2025).

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Despite this potential, existing research on educational MAS remains largely tool-centric, how learners and multi-agent systems should be optimally organised to support critical thinking remains insufficiently explored (Zheng et al., 2026).

Human–GenAI collaboration configurations

Beyond the capabilities of GenAI itself, the effectiveness of interacting with AI is influenced by how human–GenAI collaboration is configured, particularly whether learners engage with AI individually or collaboratively (Chu & Li, 2025; Tai & Chen, 2024). These collaboration configurations specify the interactional arrangements through which learners, peers and AI systems are organised within a learning environment, thereby determining how cognitive resources are distributed and how information flows among participants (McNeese et al., 2021). Accumulating empirical evidence suggests that variations in collaboration configurations are closely associated with differences in learning performance and the development of critical thinking (Fan et al., 2025; Xu et al., 2023). In particular, collaborative configurations enable learners to engage with diverse peer perspectives, which can broaden the scope of critical inquiry beyond individual reasoning (Li, Mao, et al., 2025; Zou et al., 2023). For example, learners have reported more positive perceptions of peer learning when AI-facilitated prompt sharing is incorporated, indicating that social interaction can enhance reflective engagement (Li, Mao, et al., 2025).

The integration of GenAI-powered multi-agent systems further extends and complicates human–GenAI collaboration configurations. In such environments, MAS can function as shared cognitive scaffolds that help establish a cohesive team environment and promote a collective understanding of complex problems (Zheng et al., 2026). Alternatively, they may act as autonomous third-party contributors that challenge group consensus and stimulate the re-evaluation of ideas by introducing divergent perspectives (Shi et al., 2025). Learners may engage with AI individually, collaborate with peers while receiving AI feedback, or participate in hybrid arrangements involving multiple agents and human partners. Despite this growing diversity, systematic empirical comparisons examining how learning performance and critical thinking vary across different human–MAS configurations remain scarce. Moreover, existing literature offers limited insight into the dynamic interaction processes that unfold within and across different configurations.

Interaction processes and dynamics in human–GenAI collaboration

Understanding how human–MAS collaboration supports critical thinking requires a process-oriented perspective. Critical thinking is not a static ability measured at discrete time points, but a trajectory of cognitive actions that unfolds over time. Moreover, it is not AI per se, but how AI is embedded and enacted within collaborative processes that determines learners' cognitive engagement (Díaz & Nussbaum, 2024). Gaining insight into human–GenAI interaction patterns in learning processes is therefore essential for informing the effective design of technology-integrated learning environments (Cukurova, 2025). Traditional self-report measures, while informative at an outcome level, often fail to capture the underlying 'black box' of cognitive processes, thereby necessitating more fine-grained and temporally sensitive analytical approaches (Liu et al., 2024). Recent studies have shown that the embedding of GenAI within structured tasks may influence learners' cognitive processes, with GenAI-supported groups displaying distinct cognitive engagement patterns (Kong et al., 2025; Yu et al., 2025).

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To operationalise such a process-oriented perspective, recent studies have increasingly turned to fine-grained analyses of interaction patterns, including Lag Sequential Analysis and Epistemic Network Analysis (Ba, Lu, et al., 2025; Yu et al., 2025; Zhang et al., 2025). These two approaches offer complementary insights into collaborative processes, with lag sequential analysis (LSA) capturing the temporal sequencing of interaction behaviours (Bakeman & Quera, 1995) and epistemic network analysis (ENA) modelling the co-occurrence structures of cognitive and discourse elements within interaction episodes (Shaffer et al., 2016). Their combined use enables researchers to examine both the overt dynamic flow and the implicit structural organisation of critical thinking embedded in human–GenAI collaboration (Chu et al., 2025; Liu et al., 2024). For example, Chu et al. (2025) integrated LSA and ENA to reveal differences in interaction timing across performance levels and to map divergent cognitive structures of inquiry, providing a holistic account of human–GenAI collaborative knowledge building. These approaches enable researchers to move beyond outcome measures by examining the co-occurrence and transformation of critical thinking behaviours within established process frameworks (e.g. Murphy, 2004).

Despite growing interest in multi-agent and human–GenAI learning environments, existing studies rarely integrate configurational perspectives with process-oriented analyses of critical thinking. In particular, the differential cognitive trajectories associated with individual and collaborative interactions with multi-agent systems remain underexplored (Ba, Yang, et al., 2025). Addressing this gap is essential not only for explaining the mechanisms through which critical thinking develops, but also for informing the design of AI systems that move beyond answer provision to actively support higher-order thinking.

METHODOLOGY

Research participants and context

Ninety-two undergraduate students voluntarily participated in the study at the outset. After excluding five participants due to incomplete questionnaire responses or missing final project submissions, 87 participants were retained for the final analyses. All participants were enrolled in a micro-course instructional design course at a public university located in eastern China. The sample consisted of 32 male students (36.8%) and 55 female students (63.2%), aged between 18 and 21 years. Participants represented a range of academic majors, including Preschool Education, Primary Education, Moral and Civic Education, Chinese Language and Literature, English, Physics and Geography. Prior GenAI use experience varied across participants, with the majority reporting frequent use (≥11 times per week; 55.2%). Detailed demographic information is presented in Table 1.

This study was situated in a GenAI-powered multi-agent learning environment developed to support micro-course instructional design (as shown in Figure 1). The system included four specialised agents with distinct scaffolding roles: a *Pedagogical Design Agent* supporting learning objectives, content organisation and instructional strategies; a *Technical Facilitation Agent* assisting with software operation and multimedia production; an *Ethical Reflection Agent* prompting consideration of AI limitations, academic integrity and responsible use; and a *Learning Orchestration Agent* coordinating interaction flow, turn-taking and multi-perspective synthesis.

The front-end interface of the platform was developed using Unity3D, providing an interactive virtual learning environment, while the back end integrated the Coze AI multi-agent framework with the DeepSeek large language model to enable role-based coordination and responsive collaboration among agents. Through predefined role prompts, each agent was assigned clear functional boundaries and communication styles, ensuring coherent division

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TABLE 1 Demographic information of participants ($N = 87$).

Variables	Categories	*n* (%)
Gender	Male	32 (36.8%)
	Female	55 (63.2%)
Age	18	2 (2.2%)
	19	19 (21.8%)
	20	52 (59.8%)
	21	14 (16.1%)
Major	Preschool Education	3 (3.4%)
	Primary Education	4 (4.6%)
	Moral and Civic Education	3 (3.4%)
	Chinese Language and Literature	1 (1.1%)
	English	19 (21.8%)
	Physics	28 (32.2%)
	Geography	29 (33.3%)
Prior GenAI use (per week)	Almost never (≤1 time)	3 (3.4%)
	Rarely (2–5 times)	8 (9.2%)
	Occasionally (6–10 times)	28 (32.2%)
	Frequently (≥11 times)	48 (55.2%)

次要：多代理系统介面

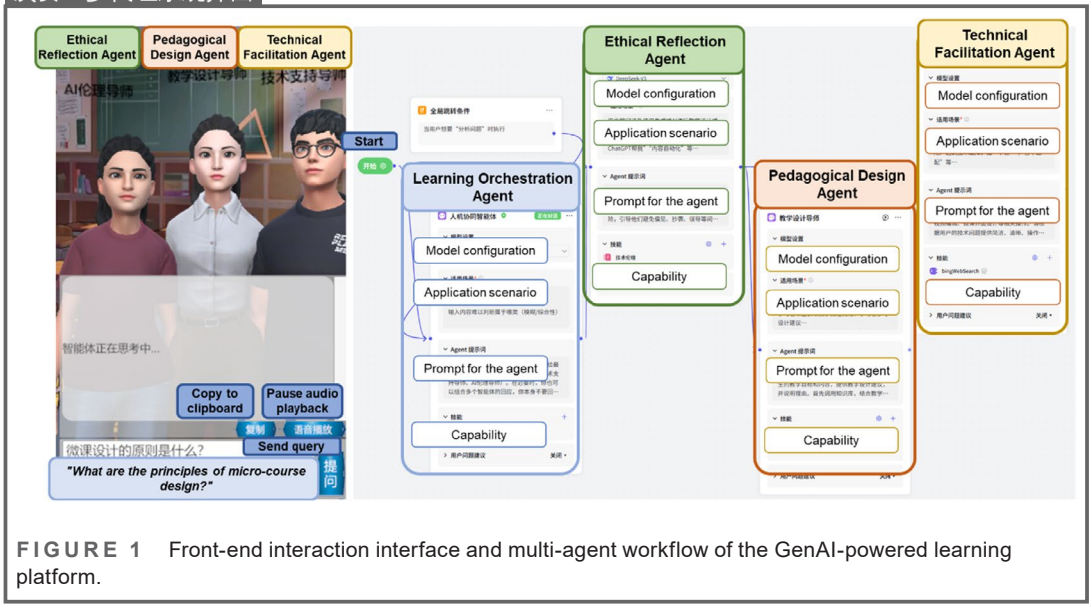


FIGURE 1 Front-end interaction interface and multi-agent workflow of the GenAI-powered learning platform.

of labour and contextualised cognitive support during the learning process. The system leveraged Coze AI's modular agent architecture, long-term memory management and knowledge base integration to sustain interaction continuity and domain relevance (Xie et al., 2025). All agents accessed domain-specific knowledge bases via a retrieval-augmented generation (RAG) mechanism, thereby enhancing the professional accuracy and contextual appropriateness of their feedback (Chen, Lin, et al., 2024). The DeepSeek model was configured with a temperature of 0.8 to balance response consistency with variability across agents.

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To support social presence and multimodal interaction, each virtual mentor was equipped with a three-dimensional avatar and text-to-speech (TTS) functionality, enabling communication through spoken language alongside visual cues such as facial expressions and gestures. These features allowed learners to interact with the multi-agent system through integrated verbal and visual channels, while also engaging with embodied interaction features such as avatars and gestures (Mayer, 2024).

Research procedure

The quasi-experimental intervention was conducted over an eight-week period within a micro-course design course. Participants were assigned to three conditions at the class level, with intact classes allocated to each condition (as illustrated in Figure 2). The three classes were regular course classes rather than selectively formed or ability-tracked groups. Although participants reported varying levels of prior GenAI use, none had prior experience interacting with a multi-agent system of the kind used in this study. In the Peer Collaboration (PC) group ($n = 26$, 13 dyads), students worked in pairs to complete the micro-course design tasks without AI support. In the Multi-Agent System Supported Individual Learning (MAS-IL) condition ($n = 33$), students individually collaborated with a multi-agent system to accomplish the same micro-course design tasks. In the Multi-Agent System Supported Collaborative Learning (MAS-CL) condition ($n = 28$, 14 dyads), students worked in pairs rather than in larger groups to minimise free-riding effects (Kao 2013), while jointly interacting with the multi-agent system to complete the design activities.

All three conditions were taught by the same instructor, who had more than 10 years of experience in instructional design and teacher education. The three groups followed identical course schedules, learning objectives, instructional content and learning resources. Pedagogical support from the instructor was equivalent across all conditions, though additional technical assistance was provided to MAS groups to facilitate platform use. The only

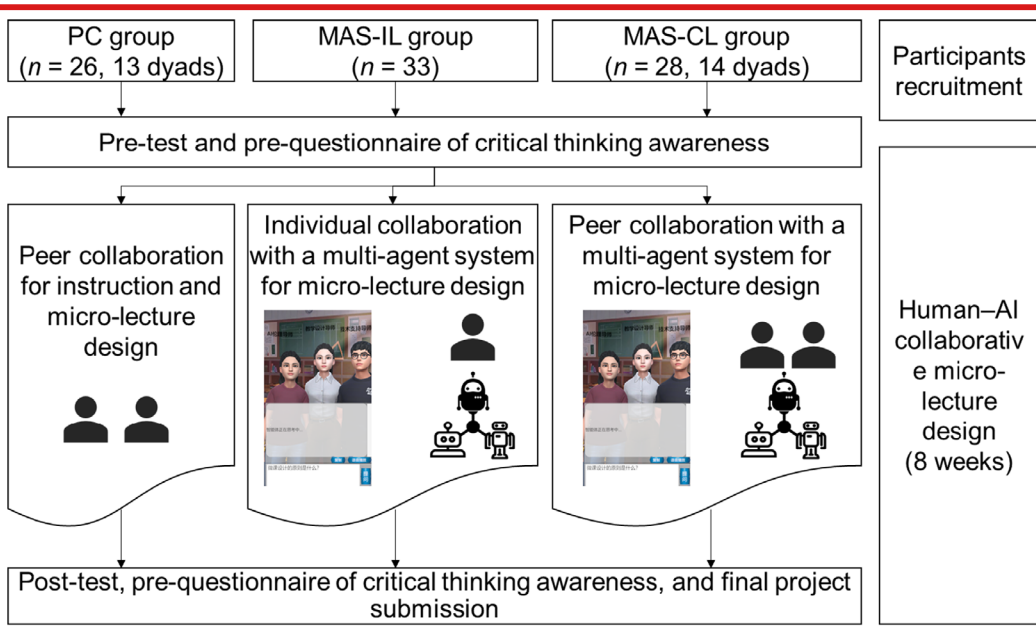


FIGURE 2 Experimental procedure.

框架：三組實驗流程

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difference across conditions lay in the presence or absence of peer collaboration and multi-agent system support.

At the beginning of the intervention, participants completed a pre-test assessing their prior knowledge of micro-course design principles and software usage, as well as a pre-questionnaire measuring critical thinking awareness. During the intervention period, students engaged in iterative design tasks, discussions and reflections aligned with their assigned learning condition. Each group participated in at least 2 hours of in-class learning activities per week. Students in the MAS-IL and MAS-CL conditions could additionally access the platform outside of class at their own discretion. At the end of the eight-week course, participants completed a post-test and the post-questionnaire on critical thinking awareness independently. In the PC and MAS-CL conditions, each dyad submitted one collaborative design project as a shared product; in the MAS-IL condition, each student submitted an individual project.

Instruments

Learning performance test

Learning performance was assessed using two complementary components to capture both conceptual understanding and applied instructional design competence: a knowledge test and a final micro-course design project. The knowledge test consisted of 10 multiple-choice items administered as both a pre-test and a post-test. The items covered key principles of micro-course design and basic software operation skills and were selected from a standardised micro-course design item bank. Two experts with more than 10 years of teaching experience in instructional design independently reviewed and refined the test items to ensure content validity. The internal consistency of the knowledge test was estimated using the Kuder–Richardson Formula 20 (KR-20), yielding coefficients of 0.617 for the pre-test and 0.788 for the post-test. Although the pre-test coefficient was modest, it was considered acceptable for a short achievement test, whereas the post-test indicated satisfactory internal consistency (Ebel & Frisbie, 1986; Kehoe, 1995).

The final micro-course design project was evaluated using an analytic rubric comprising five criteria, including instructional objectives (10%), pedagogical design (30%), technical implementation (20%), content organisation (20%) and creative design (20%), with a total score of 100 points. Detailed rubric descriptors are provided in Appendix A. Inter-rater reliability was assessed using the intraclass correlation coefficient (ICC; two-way mixed model, absolute agreement), yielding an ICC of 0.871 (95% CI [0.725, 0.939]), indicating good reliability (Koo & Li, 2016). Discrepancies between raters were resolved through discussion and consensus, and the final score for each project was determined by the consensual rating.

Critical thinking awareness scale

Critical thinking awareness was measured using an adapted scale based on Hwang et al. (2018), which has previously been applied in similar GenAI-supported learning contexts (Chu & Li, 2025). The scale employed a five-point Likert format ranging from 1 (strongly disagree) to 5 (strongly agree) and consisted of four items. A sample item is: '*When completing a task, I often pause to check whether my understanding is correct.*' In the present study, this scale was used to capture learners' self-reported critical thinking awareness during task completion, particularly reflective monitoring and evaluative checking. Cronbach's

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alpha indicated acceptable reliability at pre-test ($\alpha = 0.774$) and high internal consistency at post-test ($\alpha = 0.924$). Exploratory factor analyses (EFA) supported a unidimensional structure at both time points (pre-test: KMO = 0.733, Bartlett's $p < 0.001$; post-test: KMO = 0.849, Bartlett's $p < 0.001$), with item loadings ranging from 0.703 to 0.823 at pre-test and from 0.873 to 0.933 at post-test. The stronger post-test reliability and factor structure may reflect greater response coherence following sustained engagement with the intervention tasks, potentially compounded by repeated exposure to the same scale items (Edelsbrunner et al., 2025). The full scale is provided in Appendix B.

Critical thinking coding framework

To examine students' patterns of critical thinking processes during interaction with the multi-agent system, dialogue data were coded using a framework aligned with prior research on critical thinking processes (Chen, Zhao, et al., 2024; Zhang et al., 2025). Specifically, the coding scheme was grounded in Murphy's critical thinking framework (Murphy, 2004), comprising five phases: Recognise, Understand, Analyse, Evaluate and Create. Table 2 presents the descriptions and corresponding codes. Given that students interacted with multiple agents in turn-based dialogue, only learner-generated utterances were coded, with each interpreted in the context of the surrounding dialogue to infer the learner's underlying intent. Utterances that merely repeated or lightly paraphrased agent responses without substantive elaboration, justification or reformulation were not assigned higher-order codes. A single utterance could

TABLE 2 Coding framework.

Phases	Code	Description	Example
Recognise	R	Recognising or identifying an existing issue, dilemma, problem, etc.	How should the opening of a micro-course be designed? My topic focuses on learning to tell time
Understand	U	Exploring related evidence, knowledge, research, information and perspectives	What is the concept of a micro-course?
Analyse	A	Seeking in depth clarification, organising known information, identifying unknown information and dissecting the issue, dilemma or problem into its fundamental components	What topics can I choose? *(Follow-up)* I am designing the course for primary school students
Evaluate	E	Critiquing and judging information, knowledge or perspectives	Is the opening of a micro-course important for learning effectiveness?
Create	C	Producing new knowledge, perspectives or strategies and implementing them or acting on them	When designing a micro-course on atmospheric thermal circulation, how can the course content be integrated with music in the introductory section?
Off-task interactions	O	Utterances unrelated to the learning task or critical thinking process, including social talk or system-oriented remarks	What day is it today?

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receive more than one code where multiple critical thinking phases were simultaneously present. Two trained experts independently coded the dialogue transcripts. Inter-coder reliability was assessed through Cohen's kappa, yielding an agreement coefficient of 0.848, indicating satisfactory consistency. Disagreements were resolved through discussion.

Data analysis

Preliminary analyses indicated that the data were not fully normally distributed. Accordingly, non-parametric statistical methods were employed. Before testing the research questions, baseline comparability across the three conditions was examined. Chi-square tests were conducted for gender, age, major and prior GenAI use, while Kruskal–Wallis tests were used to examine group differences in pre-test knowledge scores and pre-test critical thinking awareness. Results indicated no significant differences across groups in age distribution, $\chi^2(6) = 4.00$, $p = 0.677$ or prior GenAI use, $\chi^2(6) = 12.16$, $p = 0.058$. However, significant differences were observed in gender distribution, $\chi^2(2) = 13.51$, $p = 0.001$ and academic major, $\chi^2(12) = 150.87$, $p < 0.001$. These baseline imbalances may be partly attributable to the intact-class assignment procedure inherent in the quasi-experimental design. In addition, no statistically significant group differences were found in pre-test knowledge scores, $\chi^2(2) = 3.73$, $p = 0.155$ or pre-test critical thinking awareness, $\chi^2(2) = 1.09$, $p = 0.580$, indicating comparable baseline levels on the two pre-intervention measures.

For RQ1 and RQ2, gain scores were calculated for the knowledge test and critical thinking awareness to capture changes across the intervention period rather than relying solely on post-test comparisons. As normality assumptions were not met, between-group differences in these gain scores were examined using the Kruskal–Wallis test. For final project performance, the peer collaboration and MAS-CL conditions involved dyadic collaboration and produced one shared project per dyad, resulting in 13 and 14 dyadic projects, respectively. To better align the unit of analysis with the collaborative structure of the task, dyad-level project scores were used for these two conditions, whereas individual project scores were retained for the MAS-IL condition. Group differences in final project scores were also examined using the Kruskal–Wallis test. Where significant effects were identified, post hoc pairwise comparisons were conducted using Dunn's test with Bonferroni correction. All statistical analyses were performed using SPSS 26.0, with $\alpha = 0.05$ as the significance threshold.

For RQ3, only learner-generated utterances were included in the process analyses, with agent turns retained only as contextual references. LSA was used to examine temporal transition patterns of critical thinking behaviours and was implemented using GSEQ version 5.1 (Bakeman & Quera, 1995). Consistent with Xu and Yang (2024), dialogue turns produced by each learner on the same day were treated as a continuous LSA sequence, on the grounds that within-day interactions constitute a coherent and temporally bounded episode. An overall Chi-square test was first used to assess whether the transition matrix deviated significantly from chance. Adjusted residuals (*z*-values) were calculated to identify statistically significant behavioural transitions, and Yule's *Q* was reported as an effect size measure for each significant transition. In addition, ENA was conducted using the WebENA tool to model the co-occurrence relationships among coded critical thinking phases across learning conditions (Shaffer et al., 2016). Following prior research on human–GenAI collaborative learning (Chu & Li, 2025), networks were constructed using a moving window of 4, normalised to account for differences in data volume, and projected into a reduced-dimensional space using singular value decomposition (SVD). A Mann–Whitney *U* test was then applied to assess significant group differences in the locations of networks within the ENA space. In line with prior research (Csanadi et al., 2018), dialogue turns were coded at

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the individual utterance level and subsequently aggregated to the condition level for the LSA and ENA analyses.

RESULTS

Learning performance

Table 3 presents the descriptive statistics and Kruskal–Wallis test results for knowledge test gain scores across the three groups. No significant group difference was observed, $\chi^2(2) = 0.825$, $p = 0.662$, indicating that all three conditions produced comparable knowledge gains over the intervention period.

Table 4 presents the group comparison results for final project scores. Project scores for the PC and MAS-CL groups were analysed at the dyad level. The analysis revealed a significant group difference, $\chi^2(2) = 9.09$, $p = 0.011$. The MAS-CL group achieved the highest mean score ($M = 89.64$, $SD = 3.67$), followed by the MAS-IL group ($M = 86.45$, $SD = 2.71$) and the PC group ($M = 85.85$, $SD = 3.65$).

As shown in Table 5, post hoc pairwise comparisons using Dunn's test with Bonferroni correction revealed that the MAS-CL group significantly outperformed both the PC group ($p = 0.039$) and the MAS-IL group ($p = 0.014$), whereas no significant difference was found between the PC and MAS-IL groups.

Critical thinking awareness

Descriptive statistics and Kruskal–Wallis test results for critical thinking awareness gain scores across the three learning conditions are presented in Table 6. The analysis revealed a statistically significant group difference in gain scores for critical thinking awareness, $\chi^2(2) = 6.298$, $p = 0.043$. The MAS-IL group demonstrated the largest gain ($M = 0.41$, $SD = 0.57$), followed by the PC group ($M = 0.06$, $SD = 1.02$), while the MAS-CL group showed a slight decline in critical thinking awareness over the intervention period ($M = -0.04$, $SD = 0.61$).

Post hoc pairwise comparisons with Bonferroni correction (Table 7) indicated a significantly larger gain score for the MAS-IL group than for the MAS-CL group ($p = 0.039$). No statistically significant differences were found between the PC and MAS-IL groups ($p = 0.473$), nor between the PC and MAS-CL groups ($p = 0.972$).

Interaction patterns

A total of 852 valid dialogue turns were generated through interactions between learners and the multi-agent system. Among these, 398 learner-initiated turns were coded and included

TABLE 3 Descriptive statistics and group comparison for knowledge test gain scores.

		Pre-test	Post-test	Gain score			
Group	***n***	***M* (*SD*)**	***M* (*SD*)**	***M* (*SD*)**	***H***	**df**	***p***
PC	26	86.15 (6.97)	90.38 (10.38)	4.23 (12.06)	0.825	2	0.662
MAS-IL	33	80.91 (17.39)	85.15 (25.26)	4.24 (21.66)			
MAS-CL	28	87.86 (7.38)	95.00 (7.93)	7.14 (10.49)			

次要：知識測驗增益（無差異）

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TABLE 4 Group comparison for final project scores (dyad-level analysis).

Group	*n*	*M*	*SD*	Mean rank	*H*	df	*p*
PC	13	85.85	3.65	26.12	9.091	2	0.011
MAS-IL	33	86.45	2.71	27.06			
MAS-CL	14	89.64	3.67	42.68			

結論：專案成績差異（RQ1）

TABLE 5 Post hoc pairwise comparisons of final project score.

Pairwise comparison	Test statistic	*SE*	*Z*	*p*
PC vs. MAS-IL group	−0.945	5.661	−0.167	1.000
PC vs. MAS-CL group	−16.563	6.658	−2.488	0.039
MAS-IL vs. MAS-CL group	−15.618	5.514	−2.833	0.014

結論：批判思考覺知增益（RQ2）

TABLE 6 Descriptive statistics and group comparison for critical thinking awareness.

		Pre-test	Post-test	Gain score			
Group	*n*	*M* (*SD*)	*M* (*SD*)	*M* (*SD*)	*H*	df	*p*
PC	26	3.94 (0.74)	4.00 (0.89)	0.06 (1.02)	6.298	2	0.043
MAS-IL	33	3.87 (0.43)	4.28 (0.52)	0.41 (0.57)			
MAS-CL	28	3.90 (0.45)	3.86 (0.59)	−0.04 (0.61)			

TABLE 7 Post hoc pairwise comparisons of critical thinking awareness.

Pairwise comparison	Test statistic	*SE*	*Z*	*p*
PC vs. MAS-IL group	−9.225	6.528	−1.413	0.473
PC vs. MAS-CL group	6.690	6.780	0.987	0.972
MAS-IL vs. MAS-CL group	15.915	6.397	2.488	0.039

in the process analyses, including 235 turns in the MAS-IL group and 163 turns in the MAS-CL group. On average, learners contributed 7.12 turns in the MAS-IL group and 5.82 turns in the MAS-CL group. Within the MAS-CL group, participation between dyad partners varied. Across the 14 dyads, the mean absolute difference in learner-initiated turns between partners was 3.92, and the less verbally active partner accounted for an average of 27.9% of the dyad's total learner-initiated turns.

Results of lag sequential analysis

All dialogue turns produced by the same learner on a given day were treated as one sequence. Across the eight-week intervention, this yielded 43 sequences in the MAS-IL group and 30 sequences in the MAS-CL group. The average sequence length was 6.44 turns for MAS-IL and 5.97 turns for MAS-CL.

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TABLE 8 LSA results for critical thinking behaviour transitions in the MAS-IL group.

Given/target	R	U	A	E	C	O
R	−0.98	−2.95	**3.71***	−0.12	1.95	−0.40
U	0.34	1.60	−0.70	−1.01	−2.00	1.19
A	1.20	−0.58	0.37	−0.94	0.83	−0.59
E	−1.20	0.20	−1.09	**2.09***	0.16	−0.50
C	0.88	0.33	−1.07	−0.84	1.25	−0.29
O	−0.71	0.33	−1.07	1.92	−0.60	−0.29

Note: Bold values with an asterisk (*) indicate meaningful transitions with adjusted residuals > 1.96 and Yule's Q > .30.

TABLE 9 LSA results for critical thinking behaviour transitions in the MAS-CL group.

Given/target	R	U	A	E	C	O
R	−1.24	−1.19	**4.46***	−1.65	−1.39	−0.14
U	−0.56	**2.66***	−1.90	−1.21	**2.90***	−3.44
A	**2.19***	−1.84	1.00	**2.02***	−1.91	−0.71
E	0.30	−1.61	−0.18	**2.17***	0.72	−0.40
C	−0.10	0.45	0.27	0.12	−0.35	−1.05
O	−1.27	0.50	−2.51	−1.69	−1.43	**8.66***

Note: Bold values with an asterisk (*) indicate meaningful transitions with adjusted residuals > 1.96 and Yule's Q > .30.

The results of the lag sequential analysis are presented in Tables 8 and 9 for the MAS-IL and MAS-CL groups, respectively. Overall sequential patterns were statistically significant for both groups (MAS-IL: $\chi^2(25) = 127.88$, $p < 0.01$; MAS-CL: $\chi^2(25) = 38.90$, $p = 0.04$). The tables report adjusted z-scores for transitions between critical thinking phases. Following established criteria, transitions with adjusted residuals greater than +1.96 and Yule's Q values above 0.30 were considered meaningful, indicating that these transitions occurred more frequently than expected by chance (Bakeman & Quera, 1995). In addition, transition cells with frequencies greater than 5 were prioritised to avoid over-interpretation of sparse transitions (Zhang & Wang, 2024). The E → E transition in the MAS-IL group (frequency = 4) was retained because it satisfied both the adjusted residual and Yule's Q criteria, although it should be interpreted with caution.

Figure 3 presents behavioural transition diagrams for each group based on statistically significant transitions. Distinct transition patterns were observed between the two groups. In the MAS-IL group, a limited number of dominant transitions were identified, most notably *Recognise* → *Analyse* (R → A) and a significant *Evaluate* → *Evaluate* (E → E) self-transition. In contrast, the MAS-CL group exhibited a broader and more distributed set of significant transitions, including bidirectional *Recognise* ↔ *Analyse* (R ↔ A), *Analyse* → *Evaluate* (A → E), and multiple transitions involving *Understand, Create* and *Off-task interaction* (e.g. U → U, U → C and O → O).

Results of epistemic network analysis

The Mann–Whitney *U* test revealed a statistically significant difference between the MAS-IL and MAS-CL groups along the horizontal axis ($U = 861.00$, $p = 0.01$, $r = -0.33$). Figure 4a,b

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次要：行為轉移圖

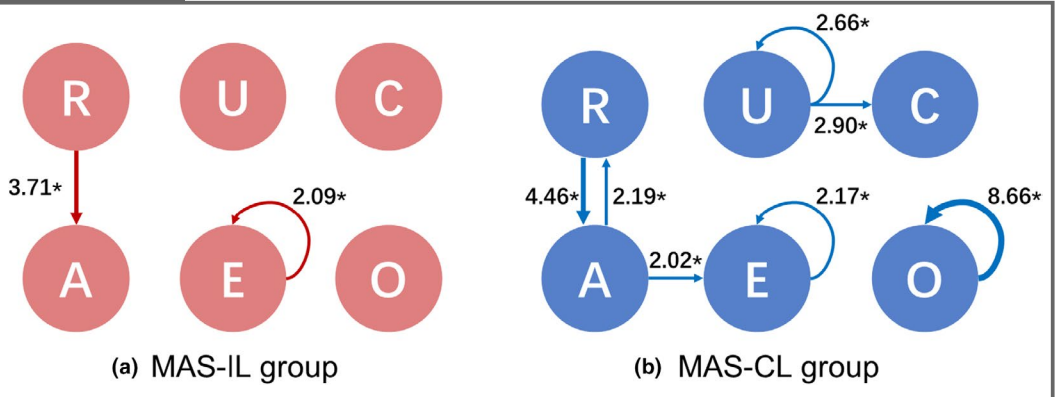


FIGURE 3 Behavioural transition diagrams in the MAS-IL and MAS-CL groups.

結論：批判思考認知網絡（RQ3）

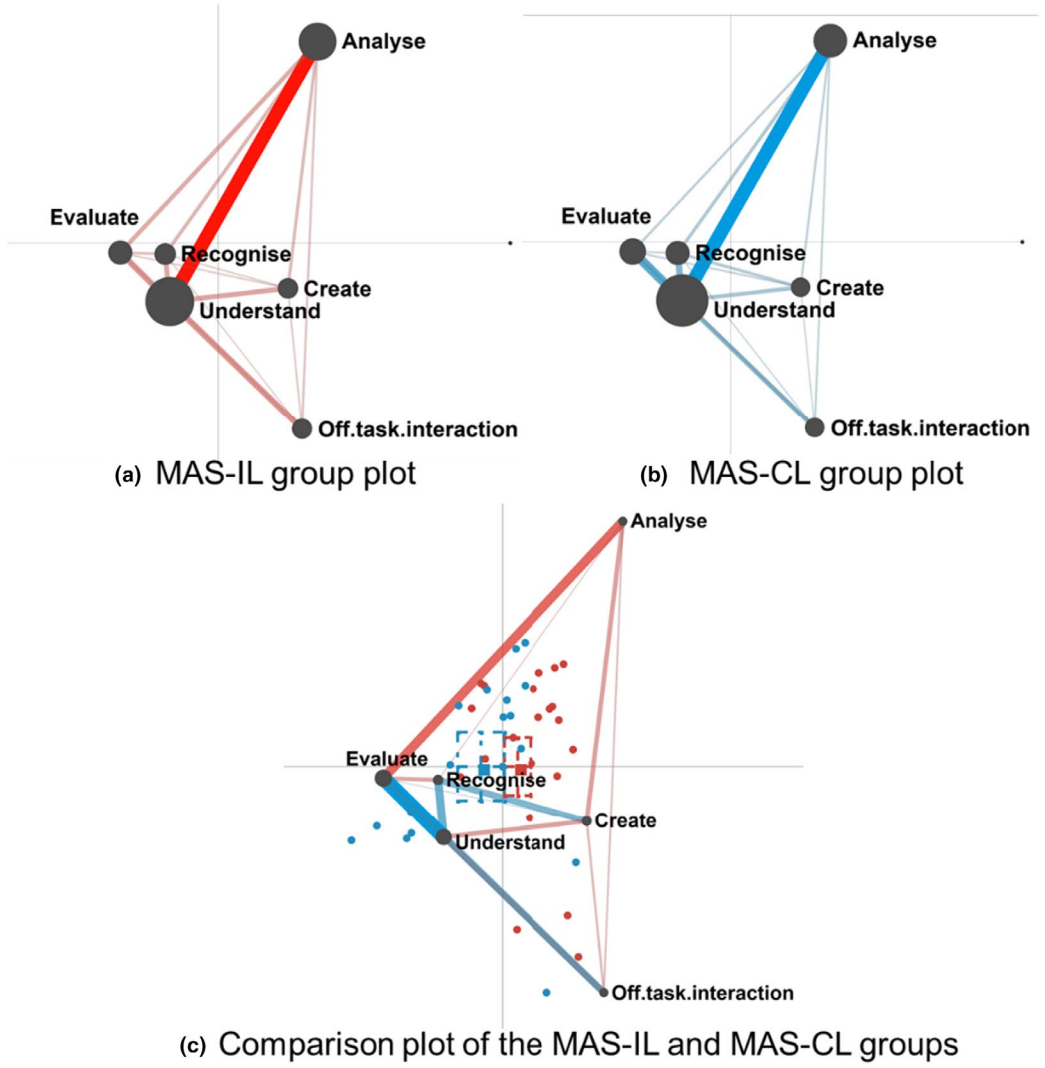


FIGURE 4 Epistemic network representations of critical thinking processes.

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present the epistemic network analysis results for the MAS-IL and MAS-CL groups, respectively. Across both groups, a strong *Recognise–Analyse* connection was observed. When the epistemic networks of the two groups were overlaid for comparison (Figure 4c), differences in edge strength became apparent, with line colour indicating the group in which a given co-occurrence relationship was stronger. In the MAS-IL group, the epistemic network was organised around a tightly connected *Analyse–Evaluate* cluster. By contrast, the MAS-CL group exhibited a more distributed and integrative network centred on *Understand*, with multiple co-occurring connections linking *Understand* with *Evaluate, Recognise* and *Off-task interaction*. In addition, a noticeable co-occurrence between *Recognise* and *Create* was observed in the MAS-CL network.

DISCUSSION

The present study examined how learning performance and critical thinking awareness differed across peer collaborative learning without AI support and two forms of human–GenAI interaction supported by a multi-agent system (MAS), and explored the critical thinking interaction patterns that emerged during human–GenAI interaction. Overall, the findings indicate that individual and collaborative engagement with a GenAI–MAS were associated with differentiated learning benefits.

Impact on learning performance (RQ1)

In terms of learning performance, no significant differences were observed in basic knowledge acquisition across the three conditions. By contrast, learners in the MAS-CL group significantly outperformed those in both the MAS-IL and PC groups in final project score. Building upon the findings of Tai and Chen (2024), who reported that both individual and paired interactions with GenAI benefited EFL learners' speaking skills, the present study further examines collaborative configurations in a multi-agent context. These findings suggest that different human–GenAI collaboration configurations may be similarly conducive to foundational knowledge and skill acquisition, while also indicating the particular value of peer collaborative interaction with a multi-agent system for more complex tasks such as instructional design and learning resource production.

One possible explanation lies in the cognitive demands associated with complex project-based tasks (Sweller, 2023). Unlike basic knowledge learning, which primarily involves remembering and understanding, instructional design projects require higher levels of cognitive engagement, including analysis, evaluation and creative synthesis (Hui, 2025). Such cognitively demanding processes can be difficult to sustain through individual information processing alone and benefit from opportunities for externalisation, negotiation and feedback. From a social constructivist perspective, knowledge construction is mediated through social interaction, negotiation and shared meaning-making (Palincsar, 2005). In this study, when learners collaboratively interacted with the multi-agent system, the presence of peers, together with role-differentiated AI agents, may have created a distributed cognitive environment that offered more comprehensive scaffolding for complex tasks, consistent with the final project rubric's multidimensional criterion, including pedagogical design and creative design. It is further reflected in the interaction pattern analyses, where learners in the MAS-supported collaborative condition exhibited significant transitions from *understanding* to *creative* synthesis, as well as epistemic networks centred on *understanding, evaluation, recognition* and *creation*.

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Impact on critical thinking awareness (RQ2)

With regard to gain scores in critical thinking awareness, MAS-IL group learners showed greater improvement than those in the collaborative MAS condition. This result suggests that individual engagement with a multi-agent system may be conducive to fostering the self-reflective and metacognitive dimensions of critical thinking. Consistent with prior studies (Li, Hwang, et al., 2025; Shang et al., 2025), the present findings indicate that, when intentionally designed, GenAI may help foster learners' critical thinking awareness, particularly its metacognitive monitoring and self-regulatory aspects. Rather than merely providing answers, MAS offered diversified perspectives and engaged learners in evidence-based reasoning, prompting them to justify, evaluate and reflect on alternative solutions (Li, Miao, et al., 2025). From a self-regulated learning perspective, interacting individually with multiple specialised agents may encourage learners to assume greater responsibility for monitoring their understanding and evaluating distinct suggestions, which constitute core antecedent awareness of critical thinking (Mohammadi et al., 2023). For example, when a MAS-IL learner asked, '*What topic can I choose?*', the instructional design mentor agent proposed several options, including cultural themes, plant-related topics and gamified learning approaches. Meanwhile, the AI ethics mentor agent added, '*When using AI-generated materials, it is important to acknowledge sources and ensure responsible use*.' In such settings, learners were positioned as the primary decision-makers who needed to actively interrogate AI-generated feedback and reconcile multi-perspective agent responses.

In contrast, learners in the MAS-CL condition showed comparatively smaller gains in critical thinking awareness. One possible explanation is that learners in this condition were required to simultaneously process the outputs of the multi-agent system and negotiate peer perspectives while meeting the demands of a complex task, which may have directed more cognitive resources towards coordination, task management and consensus-building rather than individual metacognitive monitoring, potentially constraining the resources available for deep critical evaluation of AI-generated feedback (Kirschner et al., 2018). A further plausible interpretation is that collaborative settings may attenuate individuals' perceived responsibility for sustained reflective engagement, as cognitive responsibility tends to be shared among group members. This pattern may be tentatively explained by accounts of social loafing and responsibility diffusion, which have been associated with reduced individual self-reflection in group work (Xiao et al., 2025). Behavioural evidence from the interaction data is consistent with this interpretation, as the less active partner in each dyad contributed on average only 27.9% of the dyad's total turns, indicating unequal individual engagement during collaborative human–MAS interaction. Together, these possible explanations are broadly consistent with the MAS-CL group's more distributed interaction patterns observed in the process analyses, including a higher incidence of off-task interactions. However, these interpretations remain speculative, as neither mechanism was directly examined in the present study, and further empirical investigation is needed.

Interaction patterns of critical thinking (RQ3)

Internally regulated critical thinking in the MAS-IL configuration

Beyond outcome-level differences, the combined LSA and ENA findings suggest that individual interaction with the multi-agent system supported a predominantly internally regulated mode of critical thinking. In the MAS-IL configuration, learners exhibited focused and selective interaction patterns, characterised by direct transitions from problem recognition to analytical reasoning and a tentative evaluative self-transition. Importantly, the epistemic

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network analysis further unpacked the cognitive processes underlying these seemingly linear behavioural sequences by visualising a tightly connected *Analyse–Evaluate* structure. This finding may indicate that when learners were critically engaged with the GenAI–MAS, evaluative processing may have been continuously activated beneath overt transitions between recognition and analysis, reflecting sustained internal monitoring and judgement.

Such patterns are consistent with conceptualisations of critical thinking as an internally regulated process involving iterative evaluation and refinement of ideas (Murphy, 2004). In particular, leveraging GenAI's content generation capabilities, the MAS produced multiperspective responses that resembled personalised but imperfect cases rather than definitive solutions (Guo et al., 2025). Engagement in this form of case-based discussion with the MAS appears to have more effectively fostered the evaluative dimension of critical thinking among individual learners (Zhang et al., 2025). This 'deep-dive' interaction pattern may suggest that learners treated the agents as dialectical partners, continuously evaluating information and refining ideas (Shi et al., 2025). For example, after asking the system, '*Tell me how to design a micro-lecture on uncertainty in quantum mechanics*,' one MAS-IL learner spontaneously followed up with, '*Give me some concrete knowledge points*,' indicating that the learner actively analysed and evaluated the MAS's response to intentionally deepen conceptual inquiry.

Socially distributed critical thinking in the MAS-CL configuration

In contrast, the collaborative configuration was associated with a more socially regulated and distributed mode of critical thinking. Interaction in the MAS-CL condition was characterised by reciprocal transitions between recognising and analysing, alongside iterative shifts from shared understanding to creation. Epistemic network analysis further indicated that shared understanding occupied a central position within the collaborative cognitive structure, connecting analytical, evaluative and off-task interactions. The relatively strong *Understand–Evaluate* connection may suggest that the understanding-related talk was intertwined with the appraisal and scrutiny of ideas. However, this centrality should be interpreted with caution, as it may also reflect a higher frequency of information-seeking behaviour in the collaborative setting, such as asking agents for factual clarification or task-related guidance, rather than deep conceptual processing. This may be because, in collaborative interaction with multiple agents, learners need to first establish a shared interpretive foundation before they can meaningfully evaluate and integrate diverse inputs together (Järvelä et al., 2023). From the perspective of shared mental models, such alignment of understanding is a necessary prerequisite for effective joint evaluation and knowledge co-construction in human–AI team settings (Andrews et al., 2023). Together, these process-level features suggest that critical thinking unfolded through cycles of meaning negotiation, perspective exchange and collective synthesis rather than through linear individual reasoning.

This finding aligns with prior lag sequential analysis research showing that peer-based critical thinking is dominated by analytical and evaluative processes (Zhang et al., 2025), and peer feedback in collaborative settings broadens the range of information considered and supports creative thinking (Chen, Hu, et al., 2024). For instance, during discussions on micro-lecture introductions, one MAS-CL learner innovatively proposed integrating instructional content with music, asking, '*When designing a micro-lecture on atmospheric thermodynamic circulation, how can the course content be combined with a song in the introductory section?*' This pattern of distributed cognitive engagement is further consistent with Yu et al. (2025), who found that learners using GenAI in scaffolded instructional design tasks exhibited stronger connections at higher cognitive levels, reflecting more complex and iterative cognitive processing in GenAI-integrated collaborative contexts.

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Notably, collaborative interaction was also associated with a higher frequency of off-task exchanges, such as casual questions like '*What is your name?*' and '*Why do people go to school?*'. Rather than constituting mere distraction, these interactions may reflect socially functional processes such as rapport building, coordination and conversational grounding, which are known to support collaborative sense-making and sustained engagement (Järvelä et al., 2023). In this context, critical thinking was distributed across social dialogue and human–GenAI interaction, enabling learners to co-construct understanding and generate novel ideas besides independent inquiry.

Theoretical and practical implications

This study contributes to theory by advancing a process-oriented understanding of human–GenAI collaborative learning in multi-agent environments. It extends social constructivist perspectives by broadening the locus of interaction and knowledge construction from human interlocutors to multi-agent systems (Palincsar, 2005), suggesting that the organisation of cognitive responsibility across learners, peers and AI systems may be linked to differences in critical thinking awareness and interaction processes. In addition, the findings extend theories of socially regulated learning by distinguishing between internally and socially regulated pathways of critical thinking (Järvelä et al., 2023). Individual interaction foregrounded internally regulated processes characterised by sustained analysis and self-evaluation, whereas collaborative interaction activated socially distributed sense-making and integrative reasoning. These findings deepen theoretical understanding of how human–GenAI collaboration supports higher-order thinking by explicitly linking interaction structures to underlying cognitive processes.

From a practical perspective, the findings provide actionable insights for the design and pedagogical use of GenAI-powered multi-agent systems. First, the configuration of human–MAS collaboration should be aligned with instructional goals, as collaborative engagement with multi-agent systems is particularly suitable for tasks that require sense-making and idea integration, whereas individual interaction better supports learners' critical thinking awareness and self-regulation. Second, when learners collaborate with peers and multi-agent systems, dedicated opportunities for individual reflection or explicit critical scaffolds should be incorporated to compensate for reduced individual metacognitive monitoring in collaborative settings. For example, structured frameworks that guide learners to analyse, compare and synthesise AI-generated feedback, such as those proposed by Yusuf et al. (2024), can support deeper critical engagement. Finally, AI literacy should be systematically integrated into GenAI-supported learning environments to enable learners to critically evaluate AI outputs, recognise limitations and biases and make responsible use of multi-agent support, thereby fully realising the potential of multi-agent systems for fostering critical thinking (Ng et al., 2024).

Limitations and future directions

Several limitations should be acknowledged. First, this study was conducted within a single instructional context involving undergraduate students, which may limit the generalisability of the findings to other disciplines, educational levels or collaborative arrangements. Future research across more diverse learning contexts and multi-agent system designs would help clarify the broader applicability of these findings.

Second, the quasi-experimental design constrains strong causal inference. Because participants were assigned by intact class, the conditions showed uneven distributions of

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gender and major, and the simultaneous variation in AI presence and collaboration structure further complicated interpretation of group differences. In addition, prior collaborative learning experience and prior course performance were not measured as baseline covariates, which further limits claims of group equivalence. Future research should employ fully crossed or randomised designs (Cohen et al., 2017) to disentangle the respective contributions of AI support, peer collaboration and their interaction.

Third, the scope of measurement was limited. Although the knowledge test yielded acceptable KR-20 reliability coefficients, its limited length may have constrained measurement precision, particularly in detecting subtle group differences. In addition, only overall project scores were available for analysis, precluding examination of links between specific performance dimensions and interaction patterns. Critical thinking awareness was captured by only four self-reported items, and the process analyses relied solely on dialogue data, which may not fully capture learners' actual critical thinking during interaction. Future research could incorporate performance-based assessments of critical thinking, integrate multimodal data sources such as learning analytics or physiological measures (Cukurova et al., 2020), and retain more fine-grained rubric data to strengthen the interpretation of process findings.

Fourth, the study is subject to structural limitations related to the dyadic nature of the collaborative condition. Although the knowledge test and critical thinking awareness questionnaire were completed individually, responses from learners within the same dyad may not have been fully statistically independent given their shared learning experiences. This dependence may also be associated with the lower critical thinking awareness observed in the MAS-CL group, as weaker individual accountability in dyadic interaction may have encouraged reliance on partners' cognitive contributions, thereby reducing learners' own metacognitive monitoring. In addition, dialogue turns from paired learners were tracked individually and aggregated at the condition level, and the non-independence warrants cautious interpretation of the process findings. Future research could employ multilayer network analysis approaches (Liu et al., 2026) and dyadic data analysis methods to more rigorously account for within-dyad dependence across both process and outcome measures.

CONCLUSIONS

This study investigated how learning performance, critical thinking awareness and interaction processes differed across peer collaborative learning without AI support and two forms of human–GenAI interaction with a multi-agent system. The findings indicate that the three configurations were associated with different outcome patterns. Collaborative engagement with the multi-agent system corresponded to stronger performance on instructional design tasks, whereas individual interaction was characterised by higher critical thinking awareness gains, suggesting that different configurations may afford distinct learning advantages.

Moving beyond outcome-level comparisons, this study provides process-oriented insights into how critical thinking unfolds through interaction. Individual interaction was characterised by focused, internally regulated analytical and evaluative trajectories, while collaborative interaction exhibited more distributed and integrative patterns centred on shared understanding. By integrating configurational and process-oriented perspectives, this research contributes to a more nuanced understanding of how GenAI-supported multi-agent systems can be meaningfully designed and implemented in higher education.

FUNDING INFORMATION

This work was supported by the 2026 Key Project of the Zhejiang Provincial Philosophy and Social Sciences Planning Programme (No. 26NDJC005Z), the 2024 Ministry of Education

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Country and Area Studies Project (No. 2024-N02) and the 2024 Zhejiang Provincial Education Science Planning Project (No. 2024SCG247).

CONFLICT OF INTEREST STATEMENT

All authors declare that they have no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS APPROVAL STATEMENT

The participants were protected by hiding their personal information during the research process. They knew that participation was voluntary and they could withdraw from the study at any time. Ethical approval for the study was granted by the Institutional Review Board (IRB) of Zhejiang University [2025-17] where the researchers were affiliated.

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How to cite this article: Chu, X., Dai, H., & Zhai, X. (2026). Many hands make light work? Individual and collaborative interaction with a GenAI-powered multi-agent system: Effects on learning performance and critical thinking. *British Journal of Educational Technology*, *00*, 1–25. https://doi.org/10.1111/bjet.70089

APPENDIX A

Criterion	Description
Instructional objectives (10%)	The topic is clearly defined and the content is positive and educationally appropriate, with a scientific and comprehensive presentation of the core ideas. The content aligns with course objectives and disciplinary competencies, reflecting sound educational values and pedagogical significance
Pedagogical design (30%)	The instructional topic demonstrates clear educational value and practical relevance. The knowledge structure is well organised, logically coherent and appropriately sequenced, with clear emphasis on key points. Content coverage is well-balanced and aligned with the curriculum requirements, focusing on specific knowledge points, instructional phases or experimental activities. Learning objectives are clearly stated and articulated in a standardised manner
Technical implementation (20%)	The selection and use of production tools and technologies are appropriate. Video transitions are smooth, pacing is coherent and audio-visual elements are well coordinated. The interactive design is clearly structured, with accurate navigation and functional links. Multimedia materials are applied consistently and appropriately to support the instructional theme and learning experience
Content organisation (20%)	Text, images, animations, sound effects and other multimedia elements are integrated coherently, with a well-balanced and aesthetically pleasing layout. Colour schemes are appropriate and the visual presentation demonstrates clear structure and layering. Media elements effectively highlight the instructional focus and enhance information delivery
Creative design (20%)	Originality and innovative thinking are demonstrated in the selection, processing and presentation of materials. The overall design is novel and distinctive, with creative concepts that reflect individual expression through content structure and presentation strategies
Total score	The total score is calculated as the sum of all criteria, reflecting the overall quality and comprehensive performance of the project

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APPENDIX B

1. During the learning process, I regularly reflect on whether I have achieved the goals I set earlier.
2. Before answering a question, I consider several alternative solutions.
3. When completing a task, I often pause to check whether my understanding is correct.
4. After completing a task, I ask myself how well I have performed.

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